Tidy Data: An Introduction to the Tidyverse

MA2003B Application of Multivariate Methods in Data Science

Team Members:

Raul Gomez
Adriana Reyna

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MA2003B Tidy Practice

1 Tidy Data and the Tidyverse

The Tidyverse is a collection of R packages designed for data science. It provides a consistent and user-friendly interface for data manipulation, visualization, and analysis. The core idea of behind the design of the Tidyverse is the concept of "tidy data":

Definition 1.1. Tidy Data is a way of structuring datasets to make them easier to work with. In tidy data:

1. Each variable forms a column. 2. Each observation forms a row. 3. Each type of observational unit forms a table.

This structure allows for easier data manipulation, analysis, and visualization, as it aligns with the way data is typically processed in R.

2 The WHO dataset

Before starting, we need to load the required libraries and load the WHO dataset:

```
library("tidyverse")
library("here")
library("cowplot")
library("patchwork")
library("krulRutils")
library("ISLR2")
library("magrittr")

options(scipen = 999) # Disable scientific notation
data(who)
```

We now proceed to clean the WHO dataset. The main problem with this dataset is that it contains variables (like "new_sp_m014") that are not very clear and that contain more than one piece of information. So we need to separate these variables and introduce better names for them and their associated values.

```
# We start by creating the variable "who_tidy"
#that contains the cleaned WHO dataset
who_tidy <- who %>%
  # We use pivot_longer to eliminate the variables starting with "new"
  # and use them as values instead
  pivot_longer(
    cols = starts_with("new"),
    names_to = "key",
```

```r
    values_to = "cases",
    values_drop_na = TRUE
  ) %>%
  mutate(
    key = if_else(
      startsWith(key, "newrel"),
      sub("newrel", "new_rel", key),
      key
    ),
    cases = as.integer(cases)
  ) %>%
  separate(key, into = c("new", "type", "sexage"), sep = "_") %>%
  separate(sexage, into = c("sex", "age"), sep = 1) %>%
  select(-new, -iso2, - iso3)  %T>%
  # Finally, we save the tidy data in both RDS and CSV formats
  saveRDS(here("data", "who_tidy.rds")) %>%
  write_csv(here("data", "who_tidy.csv")) %>%
  glimpse()
```

```
Rows: 76,046
Columns: 6
$ country <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", "A~
$ year    <dbl> 1997, 1997, 1997, 1997, 1997, 1997, 1997, 1997, 1997, 1997, 19~
$ type    <chr> "sp", "sp", "sp", "sp", "sp", "sp", "sp", "sp", "sp", "sp", "s~
$ sex     <chr> "m", "m", "m", "m", "m", "m", "m", "f", "f", "f", "f", "f", "f~
$ age     <chr> "014", "1524", "2534", "3544", "4554", "5564", "65", "014", "1~
$ cases   <int> 0, 10, 6, 3, 5, 2, 0, 5, 38, 36, 14, 8, 0, 1, 30, 129, 128, 90~
```

We now want to convert the columns “type”, “sex”, and “age” into factors. We start by constructing the following lookup tables:

```r
who_type_lookup_tbl <- tibble(
  code = c("ep", "rel", "sn", "sp"),
  label = c(
    "Extrapulmonary TB",
    "Relapse case",
    "Smear-Negative pulmonary TB",
    "Smear-Positive pulmonary TB"
  )
)

who_sex_lookup_tbl <- tibble(
  code = c("f", "m"),
```

```
  label = c("Female", "Male")
)

who_age_lookup_tbl <- tibble(
  code = c(
    "014",
    "1524",
    "2534",
    "3544",
    "4554",
    "5564",
    "65"
  ),
  label = c(
    "0-14",
    "15-24",
    "25-34",
    "35-44",
    "45-54",
    "55-64",
    "65+"
  )
)
```

We can now append factor columns for the corresponding variables in the WHO dataset:

```
who_factor <- who_tidy %>%
  convert_codes_to_factor(
    code_col = type,
    lookup_tbl = who_type_lookup_tbl,
    lookup_code_col = code,
    lookup_label_col = label,
  ) %>%
  convert_codes_to_factor(
    code_col = sex,
    lookup_tbl = who_sex_lookup_tbl,
    lookup_code_col = code,
    lookup_label_col = label,
  ) %>%
  convert_codes_to_factor(
    code_col = age,
    lookup_tbl = who_age_lookup_tbl,
    lookup_code_col = code,
```

```r
    lookup_label_col = label,
  ) %>%
  glimpse()
```

```
Rows: 76,046
Columns: 9
$ country     <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan"~
$ year        <dbl> 1997, 1997, 1997, 1997, 1997, 1997, 1997, 1997, 1997, 1997~
$ type        <chr> "sp", "sp", "sp", "sp", "sp", "sp", "sp", "sp", "sp", "sp"~
$ sex         <chr> "m", "m", "m", "m", "m", "m", "m", "f", "f", "f", "f", "f"~
$ age         <chr> "014", "1524", "2534", "3544", "4554", "5564", "65", "014"~
$ cases       <int> 0, 10, 6, 3, 5, 2, 0, 5, 38, 36, 14, 8, 0, 1, 30, 129, 128~
$ type_factor <fct> Smear-Positive pulmonary TB, Smear-Positive pulmonary TB, ~
$ sex_factor  <fct> Male, Male, Male, Male, Male, Male, Male, Female, Female, ~
$ age_factor  <fct> 0-14, 15-24, 25-34, 35-44, 45-54, 55-64, 65+, 0-14, 15-24,~
```

We now generate the associated plot for the cleaned WHO dataset. We want to generate a bar plot showing the number of cases grouped by type and sex. To start with, I generate a tibble that contains this information:

```r
who_type_sex_tbl <- who_factor %>%
  count(type_factor, sex_factor, wt = cases, name = "cases")
```

We can now use this tibble to generate the required plot:

```r
who_type_sex_plot <- who_type_sex_tbl %>%
  ggplot(aes(x = type_factor, y = cases, fill = sex_factor)) +
  geom_col(position = "dodge") +
  scale_fill_manual(
    values = c(
      "Female" = "#e7298a",
      "Male" = "#1f77b4"
    )
  ) +
  labs(
    title = "Tuberculosis Cases by Type and Sex",
    x = "Tuberculosis Type",
    y = "Cases",
    fill = "Sex"
  ) +
  theme_krul()
```

We can now save and plot the graph:

```r
ggsave(
  filename = here("images", "who_plot.png"),
  plot = who_type_sex_plot,
  width = 12,
  height = 6,
  dpi = 300
)
who_type_sex_plot
```

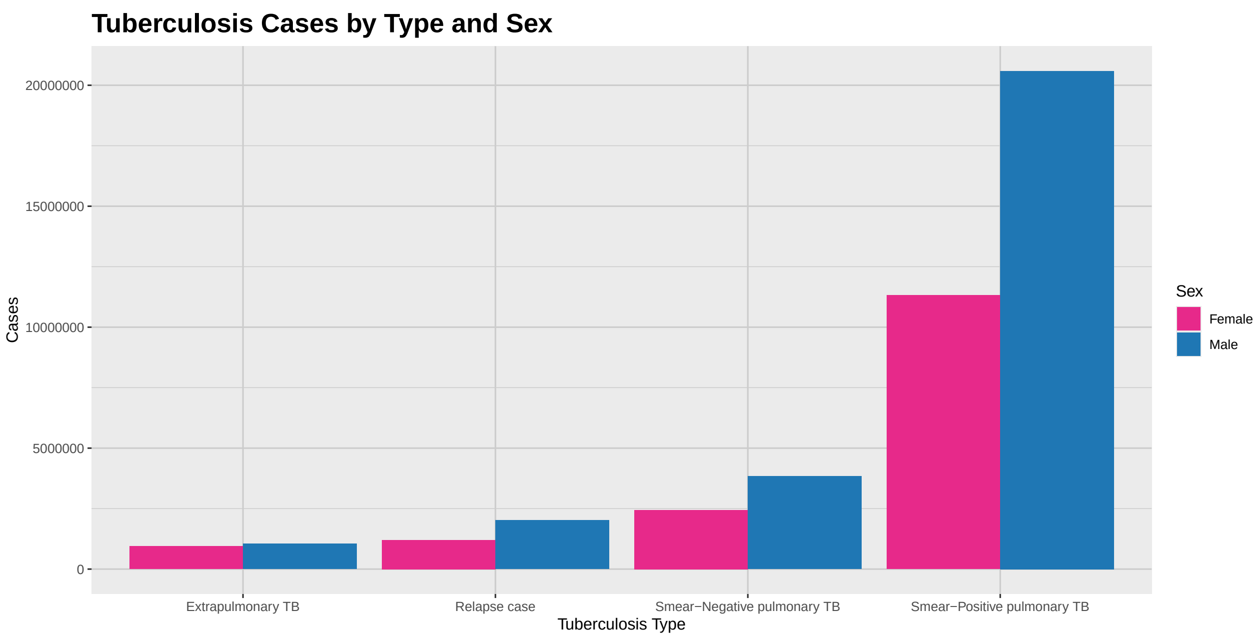

Figure 1: This plot shows a comparative analysis of the number of cases of tuberculosis by type and sex. As we can see from the plot, the number of cases is consistently higher for the male population across all types of tuberculosis.

3 The Billboard dataset

We now proceed to clean the Billboard dataset. The problem with this dataset is that the weeks in which a song appeared in the Billboard chart are represented as columns, which makes it difficult to analyze the data. So we want to move this information to a single column called "week".

```r
# We first make sure to load the Billboard dataset
data(billboard)
# Now we create the variable "billboard_tidy"
billboard_tidy <- billboard %>%
  # We use pivot_longer to move the weeks into a single column
```

```
  pivot_longer(
    cols = starts_with("wk"),
    names_to = "week",
    values_to = "ranking",
    values_drop_na = TRUE
  ) %>%
  # We now clean the "week" variable to remove the "wk" prefix
  # leaving only the week number
  mutate(
    week = if_else(
      startsWith(week, "wk"),
      sub("wk", "", week),
      week
    )
  ) %>%
  # We make sure that the "week" variable is an integer
  # We also add the "date" variable
  # which gives the date corresponding to each week
  # in which the song appeared in the Billboard chart
  mutate(
    week = as.integer(week),
    date = date.entered + weeks(week - 1)
  ) %>%
  # Now we arrange the data by date and ranking for easy reference
  arrange(date, ranking) %T>%
  # Finally, we save the tidy data in both RDS and CSV formats
  saveRDS(here("data", "billboard_tidy.rds")) %>%
  write_csv(here("data", "billboard_tidy.csv"))
```

We now want to generate a plot that keeps track of the ranking of the song “who let the dogs out”, by “Baha Men”, in the Billboard chart. We start by filtering the dataset to keep only the relevant song:

```
dogs_out_tbl <- billboard_tidy %>%
  filter(track == "Who Let The Dogs Out") %>%
  arrange(week) %>%
  select(date, ranking)
```

With this tibble, we can now generate the required plot:

```
dogs_out_plot <- dogs_out_tbl %>%
  ggplot(aes(x = date, y = ranking)) +
  geom_line(color = "#1f77b4", linewidth = 1) +
```

```
  geom_point(color = "#d62728", size = 2) +
  scale_x_date(date_labels = "%B %Y") +
  scale_y_reverse(breaks = seq(0, 100, 10)) + # rank 1 is at the top
  labs(
    title = "Chart Performance of 'Who Let The Dogs Out' by 'Baha Men'",
    x = "Date",
    y = "Billboard Hot 100 Ranking"
  ) +
  theme_krul()
```

Finally, we can save and plot the graph:

```
ggsave(
  filename = here("images", "dogs_out_plot.png"),
  plot = dogs_out_plot,
  width = 12,
  height = 6,
  dpi = 300
)

dogs_out_plot
```

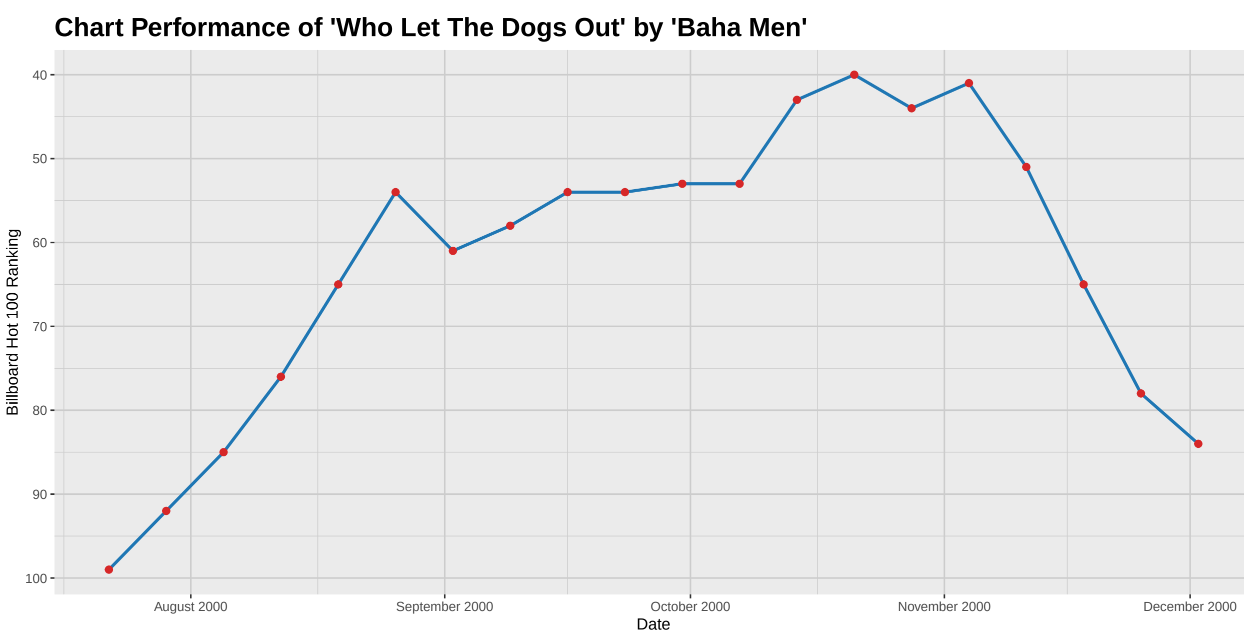

Figure 2: In this plot we can see the evolution of the ranking of the song 'Who Let The Dogs Out' by 'Baha Men' in the Billboard Hot 100 chart. The song peaked at rank 40 in the week of October 21, 2000, and remained in the top 100 for several weeks.